AP PRECALCULUS

Unit 1 • Lesson 1.5 practice • Answer key

Lesson 1.5 • Polynomial Functions and Complex Zeros

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

How many complex zeros, counted with multiplicity, does a polynomial of degree 6 have?

(A) 3
(B) 5
(C) 6
(D) 12

Solution

Answer: (C). A polynomial of degree n has exactly n complex zeros, counted with multiplicity.

Question 2

NO CALCULATOR

What are the zeros of $p(x) = x^3 + 4x$?

(A) $0,\ 2,\ -2$
(B) $0,\ 2i,\ -2i$
(C) $0,\ 4i$
(D) 0 only

Solution

Answer: (B). $x(x^2 + 4) = 0$ gives $x = 0$ or $x^2 = -4$, so $x = \pm 2i$.

Question 3

NO CALCULATOR

A polynomial with real coefficients has zeros 3 and $2i$. What is its least possible degree?

(A) 2
(B) 3
(C) 4
(D) 6

Solution

Answer: (B). $-2i$ must also be a zero, so there are at least 3 zeros.

Question 4

NO CALCULATOR

Which function is odd?

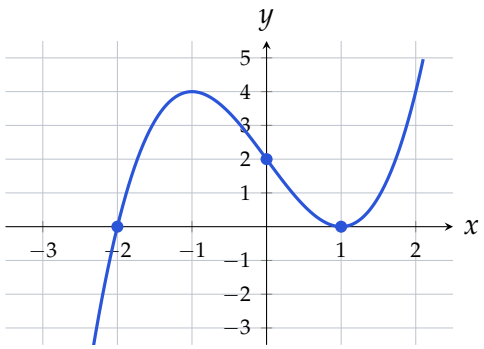
(A) $f(x) = x^4 + x$
(B) $f(x) = x^3 + 1$
(C) $f(x) = x^3 - 5x$
(D) $f(x) = |x|$

Solution
Answer: (C). $(-x)^3 - 5(-x) = -(x^3 - 5x)$, so $f(-x) = -f(x)$.

Question 5 (graph)

NO CALCULATOR

The graph of a cubic polynomial q is shown. It passes through $(0, 2)$. Which could be $q(x)$?



(A) $(x+2)(x-1)^2$
(B) $(x-2)(x+1)^2$
(C) $(x+2)^2(x-1)$
(D) $-(x+2)(x-1)^2$

Solution
Answer: (A). The graph crosses the axis at $x = -2$ (odd multiplicity), touches it at $x = 1$ (even multiplicity) and rises to the right (positive leading coefficient). Check: $(0+2)(0-1)^2 = 2$.

Question 6 (Free response)

NO CALCULATOR

Let $p(x) = (x-1)^2(x+3)(x^2+1)$.

(a) Find the degree of p and all of its complex zeros with their multiplicities.
(b) Describe the behavior of the graph of p at each x-intercept.
(c) Is p even, odd, or neither? Justify your answer.

Solution
(a) Degree $2+1+2 = 5$. Zeros: 1 (multiplicity 2), -3 (multiplicity 1), i and $-i$ (multiplicity 1 each).
(b) At $x = 1$ (even multiplicity) the graph touches the x-axis and turns back. At $x = -3$ (odd multiplicity) it crosses the axis.
(c) $p(1) = 0$ but $p(-1) = (4)(2)(2) = 16$. Since $p(-1) \neq p(1)$, p is not even; since $p(-1) \neq -p(1) = 0$, p is not odd. So p is neither.